

Monjed AI

Africa Early Warning & Action Platform — from live hazard data to a message someone can actually act on.

A warning only matters if the person receiving it can act on it. Monjed AI scores flood and earthquake risk independently, explains each in plain language, delivers it by SMS, and checks whether people on the ground can actually respond — escalating to a human volunteer when they can't.

Python (FastAPI) · MongoDB · React + Tailwind · Independent Flood & Earthquake Risk Engines · USGS + NASA POWER · Claude API · Africa's Talking SMS · 20 Countries

1 The Problem & The Idea

Most early-warning tools stop at detection: they tell a country or region that risk is rising. They rarely check whether the person receiving that warning can actually do anything about it — a blocked road, no transport, or no phone signal can make a technically correct alert useless in practice.

What most systems do

Detect risk → publish a score → send a generic alert. The chain stops there.

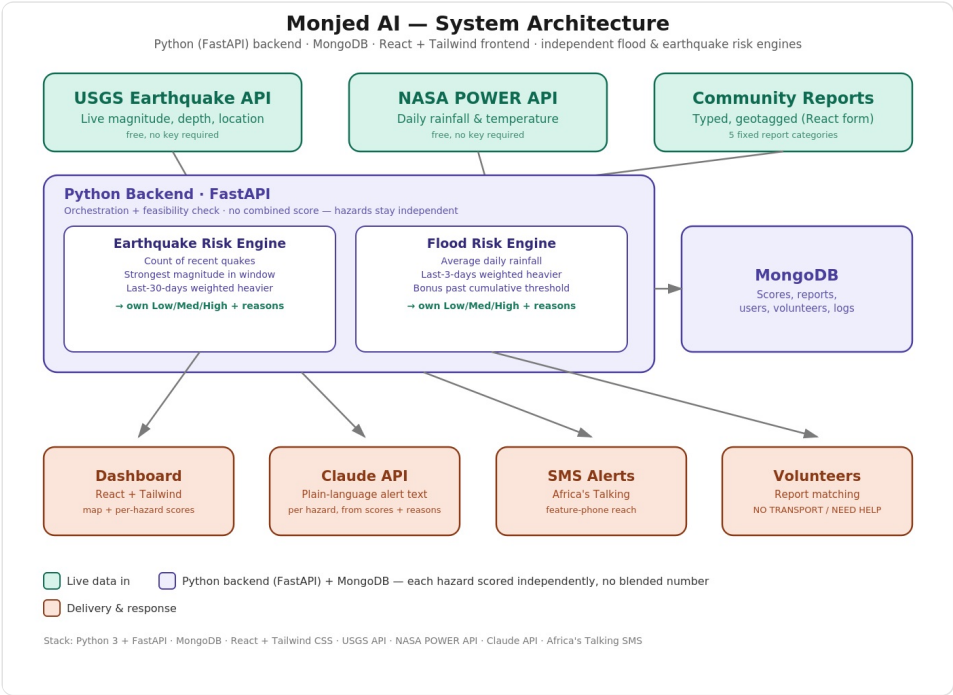
What Monjed AI does differently

Detect risk per hazard → explain it in plain language → deliver it by SMS → listen to ground reports → flag when the recommended action isn't realistic → connect the person to a volunteer.

Monjed AI combines two ideas explored during scoping: a pan-African, multi-hazard risk engine that is realistic to build in two weeks (live, free, no-approval data, each hazard scored on its own terms), and a lightweight version of an action-feasibility layer — the single most compelling idea from the Monjed concept — scoped down so it ships rather than becoming a stretch goal that never gets attempted.

2 System Architecture

A React + Tailwind frontend talks to a Python (FastAPI) backend, which runs two independent rule-based risk engines — one for earthquakes, one for floods — on top of MongoDB. Nothing in the must-build path requires a trained model, an approval-gated API, or a mobile app.



Teal = live data in · Purple = Python backend (FastAPI) + MongoDB, hazards scored independently · Coral = delivery & response

The earthquake and flood engines never write to a shared "combined score." Each writes its own 0–100 score, its own Low/Medium/High label, and its own reasons to MongoDB — the dashboard, Claude, and SMS all read from whichever hazard (or both) is active for a given country.

3 Data Foundation

Every data source is live, free, and requires no approval wait — this is what makes the two-week timeline realistic. Google Flood Hub and Copernicus GloFAS were both considered and set aside for the MVP because their pilot/access processes don't fit a two-week build; they're listed as future upgrades.

Source	Provides	Used for
USGS Earthquake API	Magnitude, location, depth, time of recent quakes	Independent earthquake risk score
NASA POWER API	Daily rainfall & temperature, near real-time	Independent flood risk score
Community Reports (React + Tailwind form)	WATER RISING · ROAD BLOCKED · NEED HELP · NO TRANSPORT · I AM SAFE	Ground-truth cross-check & feasibility signal for either hazard

Why typed reports, not free text: borrowed directly from the Monjed concept. Five fixed categories are fast for a stressed user to tap, trivial to plot and colour-code on the map, and easy to feed into both risk engines' cross-checks and the feasibility layer — without needing any NLP.

4 The Risk Engines

Not AI, not trained on anything — two transparent, **independent** weighted scoring models, one per hazard. Each ships with the reasons behind it, which matters both for building trust with judges and for what Claude needs later to write a truthful alert. The two scores are never combined: an earthquake spike in a dry country shouldn't be diluted by a low flood number, and a flood surge in a seismically quiet country shouldn't be diluted by a low earthquake number.

Earthquake score

- Count of recent quakes in-country
- Strongest magnitude in the window
- Average magnitude as a baseline
- Last-30-days weighted more heavily

Flood score

- Average daily rainfall over the period
- Last-3-days weighted more heavily
- Bonus when cumulative rainfall crosses a threshold

Output: **two separate** 0–100 scores and Low/Medium/High labels per country — one for earthquake, one for flood — each shipped as its own JSON object with its own reasons attached. Nothing is blended into a single number.

5 From Raw Data to a Message Someone Receives



One line: live data → each hazard's own risk engine → dashboard → Claude writes it in words → SMS delivers it → community reports and volunteers react to it → scores are logged per hazard for trend charts. Every step depends on the one before it, which is exactly why the build order follows this sequence.

6 Core Innovation: A Lightweight Feasibility Layer

Full personalized feasibility profiles (transport access, mobility needs, preferred language) are powerful but too heavy for two weeks — they need onboarding, per-user state, and a much bigger matching engine. Monjed AI ships a scoped-down version that captures the same judge-facing insight without the extra build time: check whether ground reports contradict the current alert, and escalate to a human when they do.

Risk alert issued

e.g. "High flood risk, Kenya" (each hazard alert is issued and evaluated on its own)

Check reports in same zone

ROAD BLOCKED / NO TRANSPORT?

Multiple matching reports?

Yes → alert flagged "verify route" + report auto-escalated in admin view.
NEED HELP / NO TRANSPORT → volunteer notified.

No / single report

Alert stands as issued — low confidence, no flag.

Confidence rule (from Monjed, kept as-is because it's simple and effective): one report = low confidence. Multiple similar reports in the same zone and time window = higher confidence and an automatic priority flag. No single report silently overrides the system.

7 Two-Way Communication & Volunteer Response

Outbound: alerts

- Dashboard: colour-coded map, always current, both hazards shown
- SMS via Africa's Talking when a country crosses medium/high threshold for either hazard
- WhatsApp channel — stretch, same message pipeline

Inbound: reports & volunteers

- React + Tailwind web form, five typed report categories, geolocation + photo
- Volunteers register with skill/vehicle tags and a zone
- NO TRANSPORT / NEED HELP reports trigger a match & notification

Minimum data stored for matching stays deliberately small: report/volunteer ID, phone, zone or approximate location, transport/mobility tag, and availability status — enough for the demo, nothing that needs a heavy privacy review.

8 Build Scope

Priority	Feature	Why
MUST	Live risk dashboard	Visual centerpiece of the demo
MUST	Independent flood & earthquake risk engines	Foundation everything else depends on — scored separately, never blended
MUST	Typed community reporting	Makes the platform interactive, not read-only
MUST	Claude-generated alert text	Highest-impact, lowest-effort AI feature
MUST	Admin / responder view	Shows a real response workflow
SHOULD	SMS alerts (Africa's Talking)	Reaches non-smartphone users; strong differentiator
SHOULD	Lightweight feasibility flag	The single most compelling idea from Monjed, scoped to ship
SHOULD	Volunteer sign-up & matching	Mostly CRUD + a distance query — realistic in time left
SHOULD	Historical trend view	Shows scores are grounded in a pattern, not one reading
STRETCH	WhatsApp channel	Same pipeline, mention in pitch
STRETCH	Offline decision capsule (PWA)	Valuable, but a full sync layer is a second project
STRETCH	Sub-national / grid-level risk	Needs finer geospatial data than the free sources provide
STRETCH	Trained ML risk model, full personalized feasibility profiles	Both need data/onboarding this timeline doesn't have

9 Two-Week Delivery Plan

Days 1-3

Data + independent risk engines

Days 4-8

Core web app: reporting, map, schema (React + Tailwind + FastAPI + MongoDB)

Days 9-11

AI + alerts + volunteers

Days 12-14

Test, rehearse, pitch

10 Live Demo Script

One scenario, start to finish — this is what judges should see, in order:

1. **Risk rises.** Open the dashboard: live USGS + NASA POWER data has pushed a country's flood score to HIGH, shown in red on the map, while its earthquake score for the same country stays LOW — the two are tracked and displayed separately.
2. **Alert goes out.** Claude's generated alert text for the flood risk appears on the dashboard and is sent by SMS via Africa's Talking to a demo phone.
3. **Ground reports come in.** Submit two ROAD BLOCKED reports and one NO TRANSPORT report near the same zone through the web form; watch them plot on the map.
4. **Feasibility flag fires.** Because reports cluster in the alert zone, the report is auto-escalated in the admin view and the flood alert is annotated "verify route before travel."
5. **Volunteer is notified.** The NO TRANSPORT report matches a registered volunteer with a vehicle in that zone; a notification fires.
6. **Admin resolves it.** Responder view shows the report moving from New → Verified → Resolved.
7. **Trend view closes the loop.** The 30-day chart shows this wasn't a single spike — rainfall has been climbing for three days, which is why the flood score crossed HIGH today, independent of the country's (unrelated) earthquake activity.

11 Limitations to State Upfront

Judges respond well to teams that are precise about what the system does and does not do. State these directly rather than waiting to be asked:

- These are rule-based risk engines, not trained predictive models — each reflects current and recent conditions for its own hazard, not a validated forecast.
- Flood and earthquake risk are scored independently and are never combined into a single blended number — a country can be High for one hazard and Low for the other at the same time. That's intentional: blending them would hide which hazard is actually driving the risk.
- Risk is computed at country level, not sub-national or grid level.
- Rainfall is sampled at one representative point per country, not averaged across its full area.
- Scoring thresholds are starting points for each engine and would need calibration against real historical disaster records before operational use.
- The feasibility layer flags contradictions between reports and alerts — it does not build a full personal profile of transport, mobility, or offline access. That's the natural next version.
- Full access to Google Flood Hub and Copernicus GloFAS requires pilot approval that wasn't available in this timeframe; USGS and NASA POWER were used instead as open, no-approval alternatives.

12 Closing Pitch

Monjed AI turns live earthquake and rainfall data into plain-language warnings — scored independently, hazard by hazard — delivers them by SMS to people without smartphones, listens to what's actually happening on the ground, and puts a human volunteer behind the moments a warning alone can't solve.

Prepared as a hackathon planning & pitch reference. Synthesized from the two-week build scope, system-pipeline walkthrough, and Monjed AI concept documents. Stack: Python (FastAPI), MongoDB, React + Tailwind. Update priorities as actual pace becomes clear after Day 3.